

paying out the maximum Social Security benefit for a couple at age sixty-sixty would cost almost \$1.2 million.³⁸

So even though U.S. pension assets amount to 70% of GDP, they should be considerably larger to fully meet pension liabilities. Either that, or pension promises are too generous. In the long run, one way or another, assets and liabilities have to be equal.

3.4. AGENCY ISSUES

The agency issues in national and defined benefit pension funds are grave. Economists and lawyers have not even settled on what the appropriate objective of a pension fund is.³⁹ This is because several parties have a claim on a private pension fund's assets, or have obligations to meet a pension fund's shortfall, and their claims often conflict. The parties include (i) beneficiaries who are current employees of the firm, (ii) retired beneficiaries who no longer work for the firm, (iii) the sponsoring firm, and (iv) the government—through the PBGC, taxpayers have a stake in all company pension plans, and the government also grants tax benefits to pension savings. Furthermore, the sharing rules are *path dependent* and *asymmetric*. If the pension fund performs well, the company may be pressured to improve benefits, which only increases its liabilities. Thus, the claims of beneficiaries increase in good times. In contrast, the cost to taxpayers through the PBGC increases during bad times.

Liability-Driven Investment

How should pension money be managed? The obvious thing would be to manage the fund to meet its liabilities. Martin Leibowitz, then an analyst at Salomon Brothers, introduced this liability-driven investment (*LDI*) framework in 1986.⁴⁰ Under LDI, the pension fund manager should manage *surplus*, which is the difference between the pension fund's asset values and liabilities, and maximize surplus returns for a given level of risk.⁴¹ The *risk aversion* incorporated in this

³⁸ As reported in "Falling Short," *Economist*, April 9, 2011.

³⁹ Technically, the economics literature has not settled on an appropriate *preference* specification (see chapter 2) of the pension fund. Love, Smith, and Wilcox (2011) deal with workers, the firm, and the PBGC. But they do not pit retired workers against current employees. Firms have great power, but not complete control, over defined pension plans. Both corporations and beneficiaries share, in different circumstances, the surplus (or deficit) of the pension plan. See Bulow and Scholes (1983), Bodie (1990a), Gold (2005), and Scherer (2005).

⁴⁰ Leibowitz, M. L., 1986, *Liability Returns: A New Perspective on Asset Allocation*, Salomon Brothers. See also Sharpe and Tint (1990).

⁴¹ Technically this is mean-variance utility (see chapter 4) over surplus. I extend this framework in Ang, Chen, and Sundaresan (2013) to incorporate downside risk.